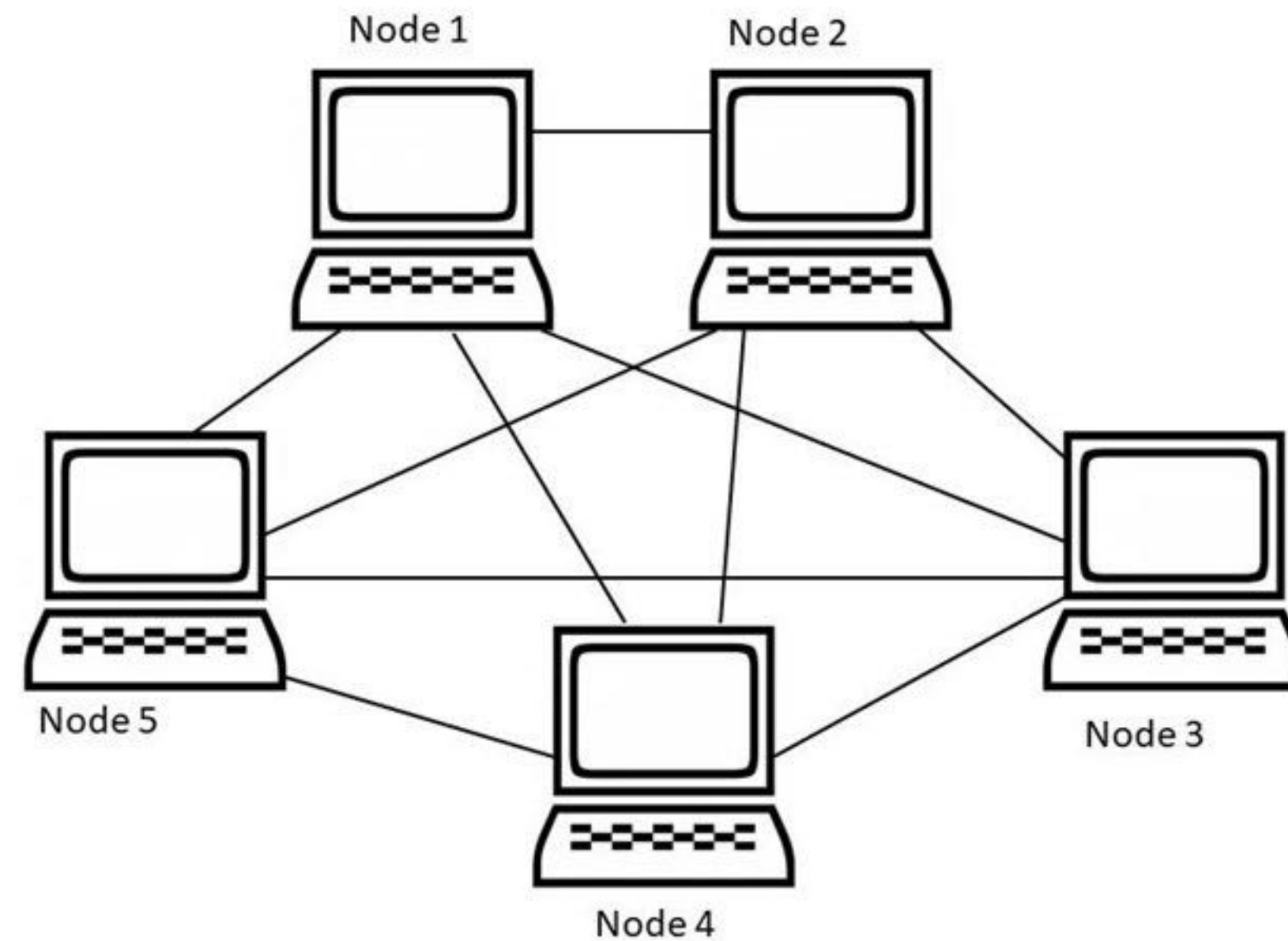


Redes HPC

Topologia Mesh

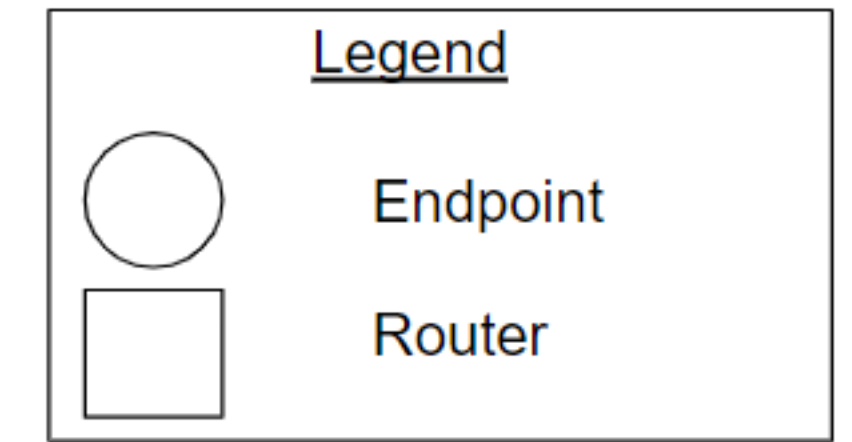
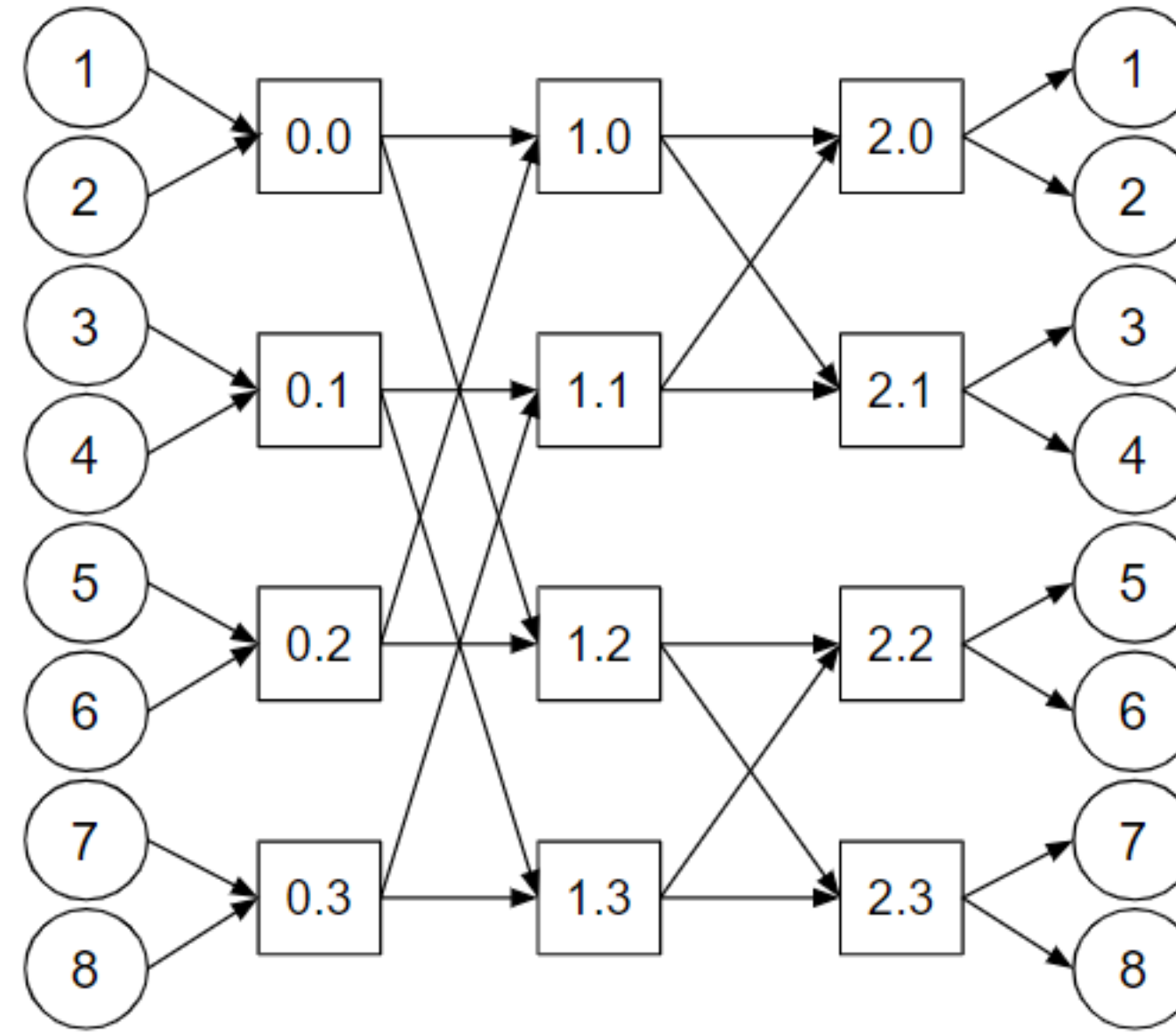
Nº de ligações

$$n (n-1)/2$$

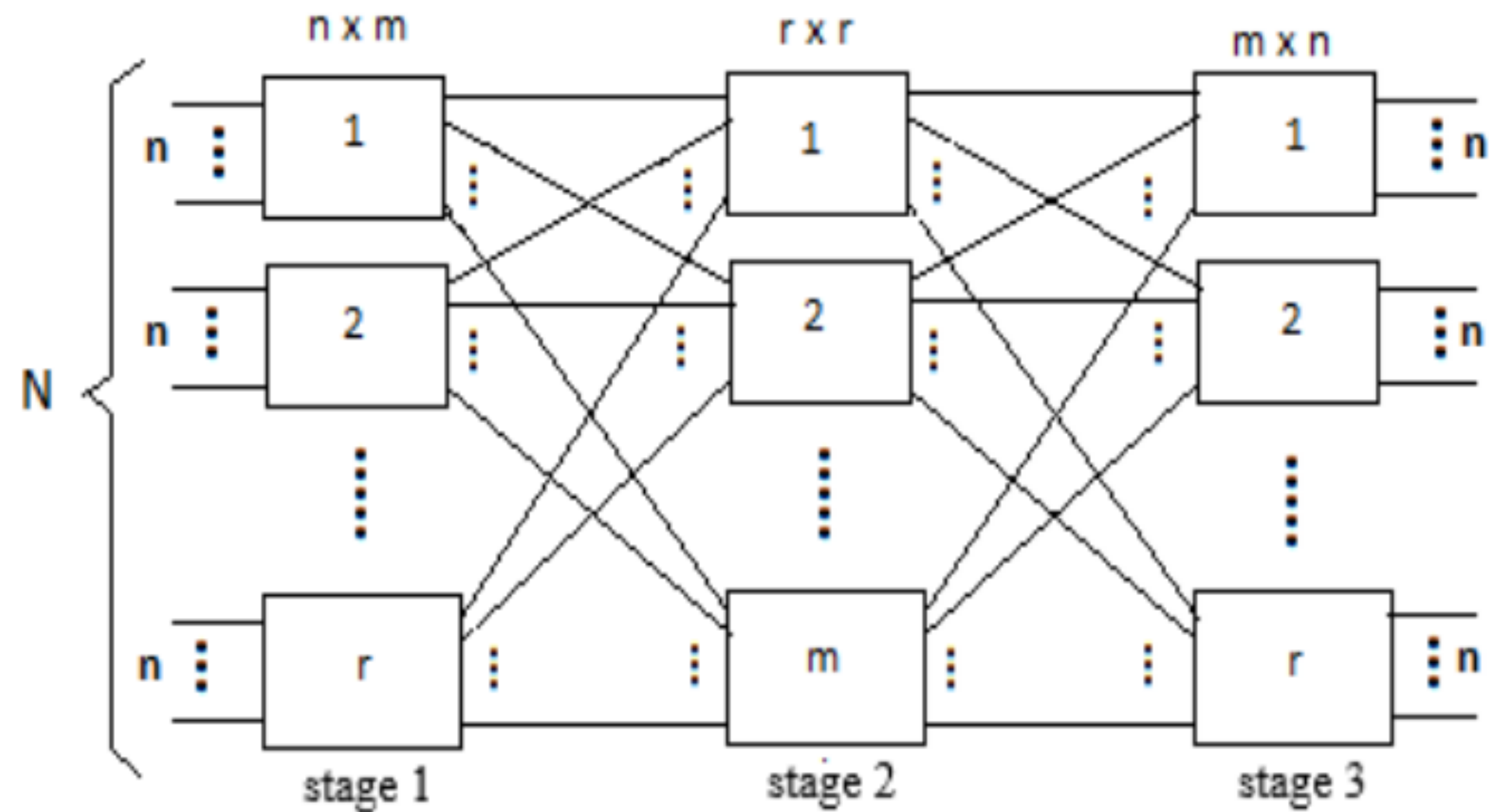


Topologia - Butterfly

- K-ary
- N-fly
- Exemplo 2-ary 3-fly

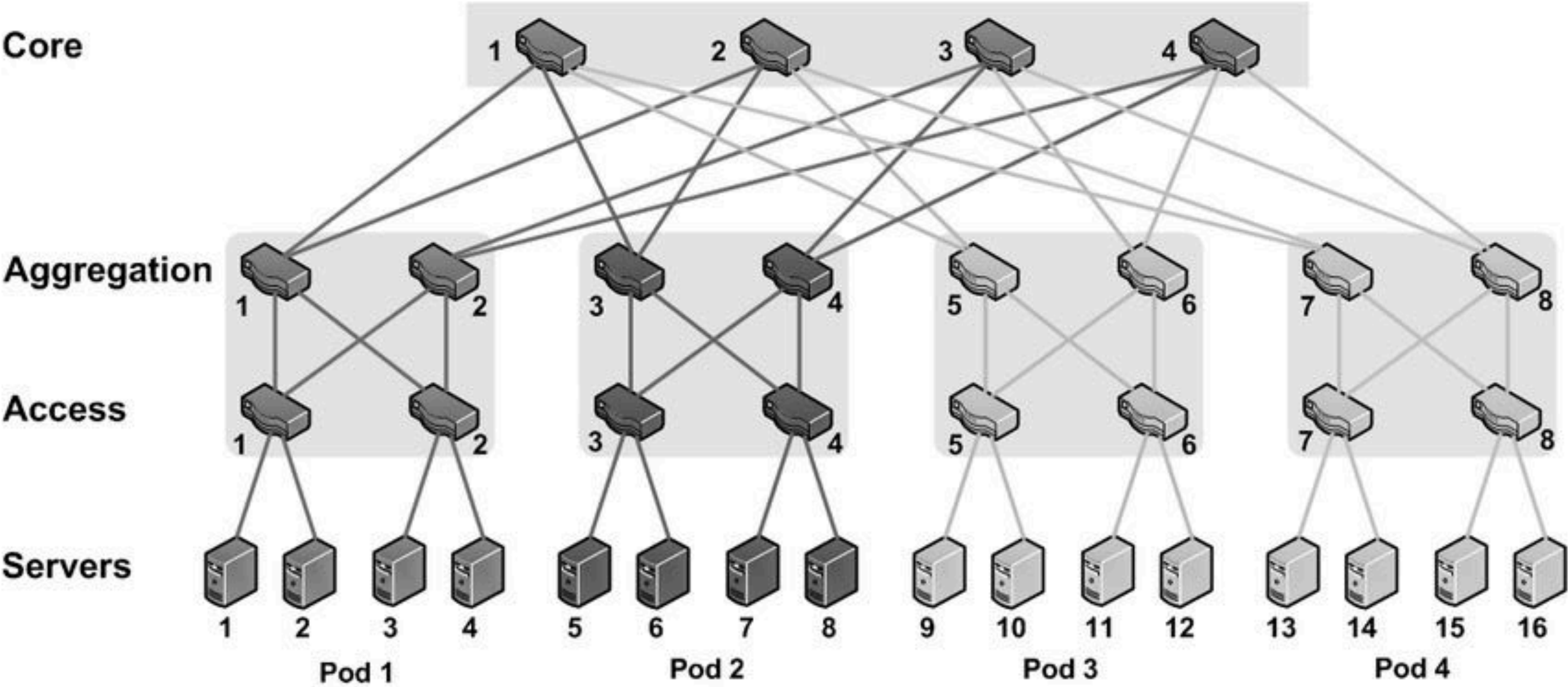


Topologia - Clos/Benes



Topologia - Fat-Tree

Parameter Name	quantity
Switch Port	k
Core Layer Switch	$k^2/4$
pod	k
Aggregation Layer Switch	$k^2/2$
Access Layer Switch	$k^2/2$
Host	$k^3/4$

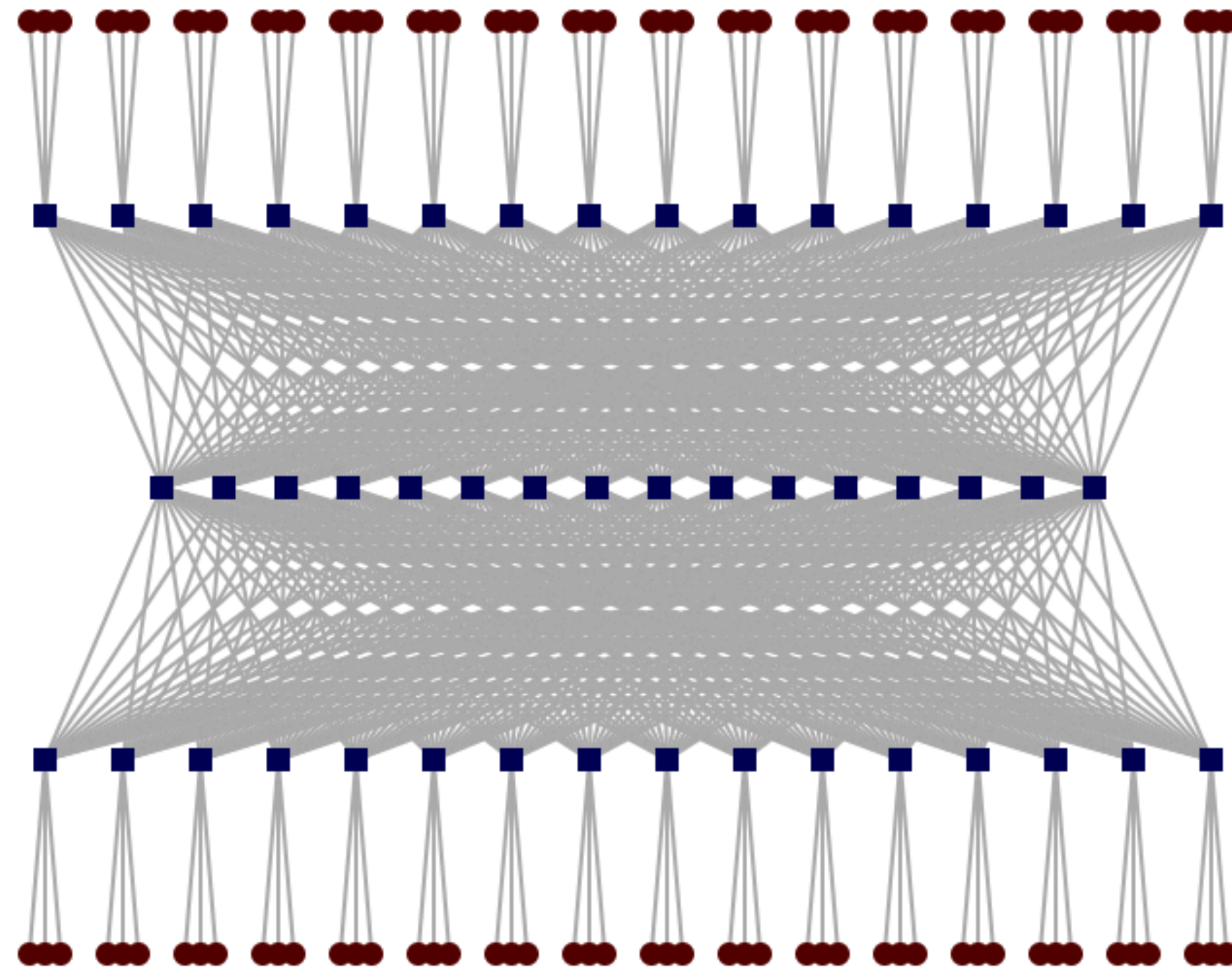


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Fat-tree

	Fat-tree with L levels	Two level Fat Tree L=2	Three Level Fat Tree L = 3	Four Level Fat Tree L=4
Number of Core switches	$\left(\frac{k}{2}\right)^{L-1}$	$\frac{k}{2}$	$\left(\frac{k}{2}\right)^2$	$\left(\frac{k}{2}\right)^3$
Number of Hosts supported	$2 \left(\frac{k}{2}\right)^L$	$2 \left(\frac{k}{2}\right)^2$	$2 \left(\frac{k}{2}\right)^3$	$2 \left(\frac{k}{2}\right)^4$
Total Switches	$(2L - 1) \left(\frac{k}{2}\right)^{L-1}$	$3 \left(\frac{k}{2}\right)$	$5 \left(\frac{k}{2}\right)^2$	$7 \left(\frac{k}{2}\right)^3$
Number of Edge Switches	$2 \left(\frac{k}{2}\right)^{L-1}$	k	$2 \left(\frac{k}{2}\right)^2$	$2 \left(\frac{k}{2}\right)^3$
Number of Pods	$2 \left(\frac{k}{2}\right)^{L-2}$	NA	k	$k^2/2$

Fat-Tree(32,3)



Topologia – Dragonfly

